Restrictions on the formation of stative passives*

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1. Stative passives, and their arguments

This paper provides an analysis of certain properties of the stative passive participle (SPP) in English: this is boldfaced in clauses like *The* ***flattened*** *boxes...*, or *The boxes are* ***flattened***. Our goal is to explain what at first glance appears to be a kind of Root-specific effect. Specifically, Roots that surface as typical unergatives like those in (1a) are deviant in the SPP; some acceptable examples with typical accomplishment verbs are given for contrast in (1b). While it looks at first glance as if the generalization to be accounted for restricts SPPs to certain Roots and not others, this appearance is misleading. There are further clauses with SPPs in which Roots that are deviant by themselves are indeed possible. This happens when they occur with Resultative Secondary Predicates (RSPs) as in (2):

(1) a. *The kitten is sneezed/laughed/coughed.
    b. The boxes are flattened/opened/emptied.

(2) Randolph's throat is laughed/sung/sneezed hoarse.

The question is thus how to derive the ungrammaticality of (1a), while permitting examples like (2) into stative passive syntax; clearly a Root-based statement like 'Roots like $\sqrt{\text{LAUGH}}$, $\sqrt{\text{SING}}$, etc. cannot appear in SPP syntax' is inadequate.

We argue that these facts fall under a larger generalization concerning the syntactic structures in which *Themes* appear. For now we are using this label for the *boxes* argument in examples like (1b), or active *Mary flattened the boxes*; we will be more precise about interpretation in §2. Put simply, our analysis holds that examples like those in (1a) are deviant because the Roots that appear there cannot be interpreted by themselves with Themes. Probing the details of how this analysis works leads to a number of questions concerning how Roots are interpreted in complex eventualities: i.e., with respect to events and states.

The questions we address derive in part from a *small* (non-phrasal) syntactic analysis of SPPs, which we have argued for elsewhere. In particular, we will assume that SPPs have

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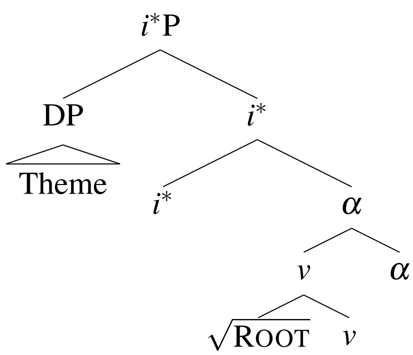
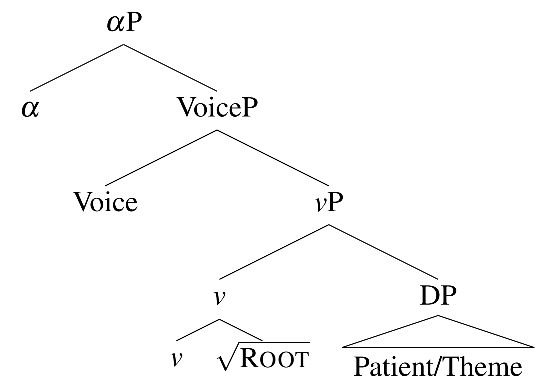
the structure in (3), where a participle-defining head α is the complement of an argument-introducing head i^* (Wood and Marantz 2017). The structure in i^*'s complement is a complex head; by way of contrast, the eventive passive (in e.g. *The boxes were flattened by the llamas*) has α with a phrasal complement, as shown in (4); (cf. Embick (2023) and Biggs and Embick (2025b)).[1]

(3) *stative passive*



(4) *eventive passive*



The question at hand is how to explain the contrast in (1) given the SPP structure in (3). The prior literature provides some context for this project. One type of analysis to be found derives stative passives from eventive passives, as stated in (5):

(5) Eventive passive ⇒ Stative passive

On this kind of approach, the contrast in (1) follows from the direction of this derivation: since an eventive passive of e.g. *sneeze* is not possible – cp. **The kitten was sneezed (by the dust)*– the input to the SPP is absent, and the deviance of (1a) is expected. A (5)-type of analysis has been formulated in both Lexicalist and syntactic theories of passive participles; see Levin and Rappaport (1986) and Bruening (2014) for representative examples of each.

A second type of approach derives (1) by direct reference to thematic relations. Anderson (1977), for example, argues that the (Lexical) rule of adjectival passive formation has this property: it produces SPPs from verbs that have Themes. The idea is that since e.g. *sneeze* does not have a Theme argument, the absence of the stative passive is expected.[2]

Assuming the syntactic structures in (3)-(4), we are not able to adopt an analysis of the type in (5): the stative passive is not built ”on” the eventive passive (or the active, for that matter). We are also not adopting a theory in which there are syntactic rules that make reference to specific thematic relations, as is required on something like Anderson (1977). At the same time, we will argue that contrasts like those in (1) derive from principles that connect directly to the Theme analysis. This point is developed in the next section, which starts with a more detailed look at stative passive interpretation.

[1]We have also argued that the SPP may appear without the i^* head under certain circumstances; see §3.

[2]Levin and Rappaport (1986) argue against the Theme analysis; we will be unable to discuss their primary points here.

2. Identifying and explaining the restrictions

We take SPPs to denote a complex change of state.[3] Illustrating with the active clause version of such a complex change, like *Mary hammered the metal flat* with a Resultative Secondary Predicate (RSP), we assume an interpretation with three eventualities: the main event e_1 (not pronounced); the Means event e_2 (*hammer* in this example), and the End State (here *flat*); see Williams (2015) for motivation and discussion:

(6) $\lambda e_1 \exists e_2 \exists s$ Means(e_1,e_2) $\wedge$ hammer(e_2) $\wedge$ End(e_1,s) $\wedge$ flat(s)

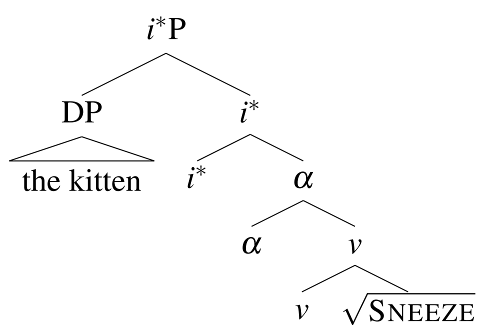
Our analysis of the SPP involves these same components, but arranged so that the state is outermost (cp. Bešlin 2023). This is shown in (7) for the SPP *flattened*. Note that we have introduced the argument of the SPP, which is interpreted as the Theme of the End state s; the PRED for the Means e_2 is unspecified, in contrast to (6) (cf. Biggs and Embick 2025a):

(7) $\lambda x\ \lambda s \exists e_1 \exists e_2$ Means(e_1,e_2) $\wedge$ PRED(e_2) $\wedge$ End(e_1,s) $\wedge$ flat(s) $\wedge$ Theme(s,x)

We take the position that that interpretations like those in (6) and (7) are produced for certain syntactic structures: specifically, those in which an eventive element is local to a stative one. Informally, the idea is that in such structures, the contextually-sensitive semantics (sometimes called *allosemy*) produces the complex change of state interpretation. Something like this has been appealed to in several domains: both for typical verbal clauses (where the highest eventuality is the clausal event), and also for certain types of stative clauses (where the highest event is stative); see Stechow (1996), Schäfer (2012), Wood (2015), Wood and Marantz (2017), and Benz (2025) (as well as §3 below).

We are assuming that SPP arguments are introduced by i^*, as in (3) above. The structure for **The kitten is sneezed* is then as follows:

(8) structure with $\sqrt{\text{SNEEZE}}$



In (8) the SPP's argument is introduced high. As noted in the introduction, given this structure it is not possible to exclude e.g. **The kitten is sneezed* on the basis of $\sqrt{\text{SNEEZE}}$ not forming an eventive passive, as the SPP is not derived from this. Instead, to explain why such examples are deviant, we will go deeper into the interpretation of SPPs. The rest of this section does this in a way that involves two components. The first concerns the requirement that SPP arguments be interpreted as Themes; this is what we call the STATIVE

[3]There is a second interpretive possibility that is discussed in §3.

PASSIVE THEME RESTRICTION, which connects in interesting ways to the prior semantic literature on changes of state. The second component, following closely on this, is that certain thematic relations may not be changed or added to. In particular, we will argue that when the change of state semantics requires that an argument be interpreted as a Theme, it is not possible to override (change or add to) this: what we call NO AGENT/THEME CHANGING/ADDING (cf. Biggs (2019)). This second component of the discussion rules out an interpretation according to which the kitten holds the End state of an event that she was the Agent of. Taken together, the two points explain the basic contrast in (1) above.

The first component of our proposal relies on the general idea that deviance results when certain Roots appear in syntactic contexts that produce a semantic incompatibility. For example, $\sqrt{\text{LAUGH}}$ appears as an unergative, but not a typical transitive ("causative"): *Mary laughed (*Sylvester)*. For reasons that will emerge as we proceed, we take this effect to stem from the fact that the syntactic structure is interpreted semantically in a way that results in deviance.

On our view, the restriction applying to SPPs is of this general type; it is stated in (9):

(9) STATIVE PASSIVE THEME RESTRICTION: Stative passive is possible only when the complement of α may be interpreted as having a Theme.

"Having a Theme" means that the material in the complement of α must be capable of being interpreted in the complex change semantics that we have employed above, irrespective of how this interpretation arises syntactically (i.e., whether in an active clause or an SPP). From the perspective of a Root, (9) enforces semantic compatibility: if the Root is capable of being interpreted with a Theme, it should be licit in the SPP; otherwise not.[4]

An important question to be addressed is how to connect the syntax and interpretation of SPPs with what happens in transitive syntax: i.e., the relation between **The kitten is sneezed* and **The dust sneezed the kitten*. We will show that this and other effects derive from the semantics for SPPs we adopted early in this section. In particular, the argument of the SPP is interpreted as the Theme of the End state. This stands in contrast to the active clause *The llamas flattened the boxes*, where *the boxes* are interpreted as the Theme of the event e_1 (cf. Williams 2015). The question is to how to link the latter two observations in a way that can derive (9). Specifically, what is needed is a way of encoding the fact that SPPs are impossible when the complement of α does not take a Theme in active syntax; bearing in mind that active syntax is **not** part of the SPP on our analysis.

It turns out that the link in question has been discussed in the literature on complex changes. Parsons (1990) in particular hypothesizes that these eventualities display a bidirectional relationship connecting e_1 and its End state s: the Theme of e_1 is necessarily the Theme of s, and vice versa. In the notation adopted here, this is (10):[5]

(10) $\text{End}(e_1,s) \rightarrow \text{Theme}(e_1,x) \equiv \text{Theme}(s,x)$

[4]Other syntactic contexts beyond those considered here are plausibly understood as requiring a Theme interpretation for their argument: e.g. "Middles" (*These boxes flatten easily* vs. **This kitten sneezes easily*), or *-able* adjectives (*This box is flattenable*, **This kitten is sneezable*).

[5]See Parsons (1990:119), who uses the relation BECOME rather than End: "... the Theme of [BECOME's] event is the same as the Theme of its target state." Our formulation in the text is based instead on (6) above.

In the case of active *flatten the boxes*, (10) ensures that the boxes are the Theme of the End state s. In the SPP, the argument introduced by i^* is a Theme (state holder) of the End state s, and linked by (10) so that it is also interpreted as the Theme of event e_1.

The next question to ask is how these interpretations relate to various types of complements for α in the SPP. In the case of e.g. $\sqrt{\text{SNEEZE}}$, it appears to be the case quite generally that this Root does not appear by itself in structures with Theme arguments: cf. **The dust sneezed the kitten*, or perhaps **Mary sneezed Susan*. The deviance of **The kitten is sneezed* follows accordingly: in brief, the syntax forces an interpretation in which *the kitten* must be understood as a Theme argument; but this Root by itself is incompatible with interpretations that require it to be understood in such eventualities.[6]

In addition to what we have just seen, there is a second reason why *sneezed* is deviant in the SPP, which has to do with how End states are defined. This point can be developed by considering Roots that appear in both unergative and transitive syntax, such as $\sqrt{\text{SING}}$. The situation with these is more complex than with e.g. $\sqrt{\text{SNEEZE}}$. $\sqrt{\text{SING}}$ can clearly be intransitive– *Cynthia sang*– or transitive, as in (11). At issue are the SPPs (11a,b):

(11) Cynthia sang the aria.

a. *Cynthia is sung.

b. #The aria is sung.

While (11a) is deviant, it is possible for (11b) to be acceptable in particular contexts: what is referred to elsewhere as a *Job is Done* (JiD) interpretation. In a situation in which the singing of the aria is an item on a list to be checked off, (11b) can be uttered felicitously: e.g., *Ok, the aria is sung, we can go home now.* As discussed in Biggs and Embick (2025a), JiD is a type of stative interpretation in which the context provides an End state for a Root that typically does not occur with one. It appears that at least this component of the interpretation of the SPP is potentially subject to substantial coercion.

In another type of example introduced above, an End state is not supplied by coercion, but by the syntax. This is the case with the examples with RSPs. With these, the idea is that the semantics introduces a complex change of state of the type that has a Theme argument, as in (6). So, while e.g. $\sqrt{\text{LAUGH}}$ does not appear with a Theme argument on its own, $\sqrt{\text{LAUGH}}$ with an RSP like in *laugh hoarse* does occur with a Theme. The acceptability of the SPP in *Mary's throat is laughed hoarse* is thus expected (cp. *Mary laughed her throat hoarse*). As we noted in the introduction, the generalization that is at play is not about Roots per se, but about the complement of α, which might contain material in addition to the Root. On this last point, there is a crucial question concerning how the relevant interpretation is produced in *laughed hoarse*; i.e., what syntactic structure is found when SPPs and RSPs are found together. We address this in §3.

Moving ahead to the second component of the analysis, the question to be considered concerns what **The kitten is sneezed* might conceivably mean. One possibility– the one that we have ruled out above– has the kitten as the Theme of the sneezing. Another possibility,

[6]It is conceivable that something like an "accomplishment" interpretation for e.g. $\sqrt{\text{SNEEZE}}$ could be coerced under certain circumstances, where the context provides a suitable end state (cf. Biggs and Embick 2025a). If so, then our prediction is that the SPP should be possible on precisely this interpretation.

to be excluded, introduces Agentivity: specifically, it must be explained why **The kitten is sneezed* cannot mean that the kitten holds an End state of a sneezing event of which she was the Agent (instead of or in addition to being a Theme).

There are at least two reasons to consider the latter possibility. The first is that part of SPP interpretation is subject to strong coercive effects; we saw this earlier in connection with the JiD meaning of *The aria is sung*. Effects of context/coercion could in principle apply in the case under consideration as well, where the kitten could conceivably be thought of as holding the End state of a sneezing that she did. The second reason is that examples in which an argument is interpreted both as the Agent and the Theme of a complex change are indeed attested; this is what happens with binding in examples like (12a).[7] Crucially, the SPP cannot have this interpretation (12b):

(12) a. Wanda smoked herself thin.
b. *Wanda is smoked thin.

The deviance of the SPP (12b) should follow from the same principles ruling out **The kitten is sneezed*; we will now show how.

To start with, we note that the active (12a) means that Wanda became thin by virtue of a complex change of which she was the Agent; the Means of the change is smoking. *Smoking* here is understood such that the undergoers of the the smoking event– e.g. cigarettes, or another tobacco product– are not expressed.

This last point is important because there is indeed a grammatical interpretation of (12b); viz., one in which Wanda is understood as the Theme of a change. Crucially, the latter interpretation is possible in the SPP:

(13) Wanda is smoked thin.
a. *Int. 1: became thin by smoking tobacco
b. Int. 2: became thin by having been inserted into a smoker[8]

The availability of Int. 2 (13b) helps to establish our basic point. In examples like **The kitten is sneezed* or **Wanda is smoked thin* (Int. 1), the argument is merged as the specifier of i^*, as shown above. Its interpretation in this position is as a Theme– the holder of the state variable; by meaning postulate it must also be understood as the Theme of the event that ends in that state. This is incompatible with Int. 1, which crucially requires that Wanda **not** be understood as a Theme, i.e. as having been smoked.

The patterns under consideration suggest that the argument introduced by i^* must be interpreted as a Theme, and nothing more: there is no possibility to change it into another thematic relation like Agent, or adding this role to produce a ”dual role” interpretation of the type produced with binding in *Wanda smoked herself thin.* The basic principle that at play appears to be something like the following:[9]

[7]The same can happen in typical reflexives like *Mary pounded herself flat* (in a cartoon, for example).

[8]This is a cooking device related to the barbecue.

[9]We intend this principle to cover the assignment of thematic relations. There is more to be said about possible ”back door” mechanisms that might be at play in certain SPPs (or eventive passives, for that matter).

(14) NO AGENT/THEME ADDING/CHANGING: An argument interpreted as holding the Theme relation may not be coerced/contextually affected to bear the Agent relation (and vice versa).

The statement in (14) is restricted to Themes and Agents because that is what is required for the case study at hand. In addition to the role it plays in SPP interpretation, it has some more general applications; for example, it rules out "reverse" mappings, where e.g. *The wall VERBED Mary* could have the wall as Theme and Mary as Agent for a particular Root. The intention behind including "adding" in (14) is rule out scenarios in which a single argument that is a Theme also acquires an Agent relation beyond this (cf. (13) above). We will consider how general (14) is along other possible dimensions in the conclusion.

To summarize the main points of this section, our account of restrictions on SPP formation is in part syntactic, and in part semantic. The small syntax of the SPP results in its argument being introduced externally, by i^*. The structure produces the semantics of a complex change, which requires in turn that the SPP's argument be interpreted as a Theme; there is no way to override (or add to) this interpretation contextually. When the complement of α contains material that cannot be interpreted with a Theme, such as when e.g. $\sqrt{\text{SNEEZE}}$ or $\sqrt{\text{LAUGH}}$ appears by itself, the SPP is deviant.

3. A note on SPPs and RSPs

As we have seen, SPPs can appear with RSPs; both when the object is interpreted as undergoing the action defined by the Means (15), and when it is not (16):

(15) a. Mary hammered the metal flat.
b. The metal is hammered flat.

(16) a. Om ran the soles of his sandals thin.
b. The soles of Om's sandals are run thin.

The question we turn to in this section is how the SPP and the RSP combine in these examples to produce a Theme interpretation for the argument introduced by i^*. The point to focus on concerns how the eventualities to which the Theme is related should be analyzed. In a typical verbal clause like *hammer the metal flat*, the event e_1 is highest; in an SPP like *the metal is hammered flat*, the state s is highest. How can the semantics of these two situations be unified?[10]

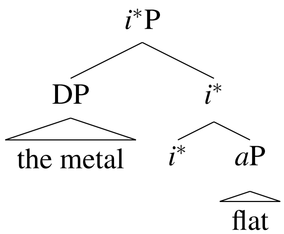
We believe that a satisfactory answer to this question is produced by an analysis in which (i) the basic syntax of the (15b)-(16b) examples is one in which the participle is attached to an *a*P, which is the RSP, and (ii) the contextual semantics operates exactly as it does in active clauses. In particular, for the second point we will make crucial use of the idea from the beginning of §2 that the complex change of state semantics is produced in the semantics when there is an eventive element local to a state one.

For example, it is possible to understand *John is painted* as involving John as the painter. Our claim is that this effect is not due to thematic relations– i.e., assignment of the Agent role to John– but instead arises in the pragmatics.

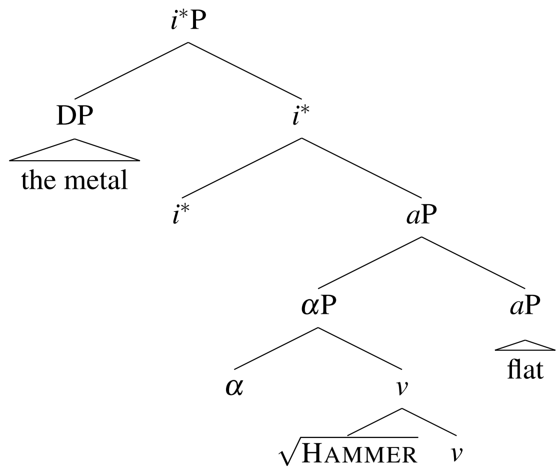
[10]See Embick (2023) for additional context on this question.

Suppose first that with standard adjective phrases (*a*Ps) like *flat*, the argument is introduced by i^* (this is a version of the view according to which adjectives are "external" predications); and that with *hammered flat*, *hammered* is adjoined to the *a*P:

(17) *The metal is flat.*



(18) *The metal is hammered flat.*



The argument in both (17) and (18) is introduced by the adjective's i^*. The question then is how the complex change semantics is produced in (18) (cf. Embick 2023).

The way in which we propose to do this relies crucially on an aspect of (18) that has not been seen to this point. In this structure, the participle is a "bare" αP, without an i^* of its own. The use of αP without i^* here follows a proposal that we justify at length in Biggs and Embick (2025b). There, we demonstrate that SPPs have two distinct interpretations: one stative, which we have been dealing with in earlier pages; and one that is purely eventive. The purely eventive reading is seen typically in attributive syntax: see e.g. *the secretly filmed wombats*. Importantly, the effect is argued to be contextual, with the eventive interpretation arising when α is not the complement of i^*. On the assumption that the participial αP is eventive in this way, the semantics of a complex change of state is produced via the contextually-defined rule that is first seen in §2 above, around (6).

For the specific cases under consideration, Benz (2025) provides a number of arguments showing that the syntax produces a constituent in which an RSP is the complement of the verb: e.g. [$_v$P [$_v$ $\sqrt{\text{HAMMER}}$ v] [$_a$P flat]]; a so-called "complex predicate" analysis. Our proposal is that although the relative height of the eventive and stative components differs in this structure and SPPs like (18), the context-determined semantics of change is produced for the same reasons in each case: viz., because each consists of an event and a local state. On the one hand, this proposal requires a certain amount of flexibility in the semantics, as the end result is a state in one case, and an event in the other. But the benefits that accrue from this– potential unification of resultative semantics across categories– suggest that this is a move worth making (cf. Benz 2025 for discussion).

4. Conclusions

We close by noting some implications of our analysis; first, concerning the Theme Restriction; and second, concerning the idea that Themes may not be converted into Agents.

On the first front, it is useful to observe that the STATIVE PASSIVE THEME RESTRICTION explains (or in one sense recasts) a generalization that has been widely discussed in the literature: the *Sole Complement Generalization* (SCG), which observes that SPPs are possible for arguments that may appear in a monotransitive (Levin and Rappaport 1986). Our explanation for this effect relies crucially on the semantics of complex changes, and the entailments that arise in these: specifically, that the Theme of a state is the Theme of the event that ends in that state, and vice versa. Crucially, we are able to explain patterns that cross categorial environments (typical verbal clauses; SPPs) due to the idea that the semantics of complex changes of state is found with both.

This analysis holds that SPP's are restricted in a way that is basically semantic: it relies on whether or not an interpretation is possible, and not on a syntactic notion like transitivity, or direction of derivation. At the same time, there is something fundamentally syntactic about the approach, due to the way in which the semantics of complex changes is produced when certain structures are interpreted. Taken at face value, our analysis requires that the same Thematic relation be associated with more than one structural position: it has the Theme relation assigned both to the argument introduced by i^* in the SPP in e.g. *The box is flattened*, and to a ”typical” *v*P argument in e.g. *Mary flattened the box*. One further question to investigate is the precise internal structure of the *v*P internal case: there are some different possibilities here that would be interesting to explore in future work.

Turning now to contextual effects, there are interesting points to be made on NO AGENT/THEME ADDING/CHANGING (NATAC; (14)) in connection with some other proposals concerning contextual interpretation. For the specific case of participles, apparent ”voice reversals” have been reported in the literature; e.g. *confessed killer*, or *escaped criminal*; see Bresnan (1996) and related work. The English examples she adduces appear to be of different types; we do not have the space/time to develop this point here, however. Directly related to our main claims, Paparounas (2025) analyzes the Greek equivalent of *John is eaten*, which can mean either that John has been eaten by someone/something; or that John is in the end state of an eating event of which he was the Agent. As he discusses, certain ways of analyzing this effect are compliant with NATAC, while others are not.

On the question of possible extensions of NATAC, we framed this restriction in §3 as specific to these two relations, because it is not clear how general it is. It has been proposed that there are indeed certain contextual effects in which thematic relations are assigned to an argument. For instance, Wood and Marantz (2017) have external arguments introduced in a generalized way by i^*, such that the particular Thematic relation that is associated with this head's argument is defined by its local context. In this scenario, notably, there is no *Root Override* of typically assigned relations (cf. Biggs (2019)), making it plausibly different in kind from the kind of effect covered by NATAC. Another question for future research thus concerns whether the kinds of contextual effects arising when functional morphemes are interpreted are different in kind from those involving Roots. In short, as allosemy is appealed to more and more in current analyses, it must be asked what its limits are.

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